Program : **Diploma in Robotic Process Automation**	
Course Code : **6309C**	Course Title: **Artificial Intelligence Lab**
Semester :**6**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3(L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- Become familiar with basic principles of AI towards problem solving, knowledge representation and learning.

Course Outcomes:

On completion of the course student will be able to:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Study of Prolog.	6	Applying
CO2	Preparing Symbolic Language	9	Applying
CO3	Facts, Rules and Queries representation	12	Applying
CO4	Implement AI problems	15	Applying
	Series Test	3	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	2	3	3		
CO2	3	2	2	3	3		
CO3	3	2	2	3	3		
CO4	3	2	2	3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Study of Prolog.**		
M1.01	Basics of Prolog. a. Learn Prolog programming language basics. b. Implement basic sample programs in Prolog.	6	Applying
CO2	**Preparing Symbolic Language**		
M2.01	Learn how to represent facts using Prolog	4	Applying
M2.02	Write sample program to represent facts using Prolog	5	Applying
	Series Test – I	1.5	
CO3	**Facts, Rules and Queries representation**		
M3.01	Implement programs to represent facts using propositional logic	4	Applying
M3.02	Implement programs to represent facts using First-order logic	4	Applying
M3.03	Write other conditional programs using Prolog	4	Applying
CO4	**Implement AI problems**		
M4.01	Implement sample Toy problems in AI	5	Applying
M4.02	Implement sample Real-world problems in AI	5	Applying
M4.01	Build advanced AI problems.	5	Applying
	Series Test – II	1.5	

**** - Micro Project**

Students are expected to do a micro project in Artificial Intelligence during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record. Example: Develop program such as

- **Stock prediction**
- **Chatbot**

- **Social media suggestion**

Text / Reference

T/R	Book Title/Author
T1	Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd Edition. Prentice Hall.
T2	Nilsson N.J., Artificial Intelligence - A New Synthesis, Harcourt Asia Pvt. Ltd.

Online Resources

Sl.No	Website Link
1	https://www.academia.edu
2	https://www.studocu.com/
3	https://github.com
4	https://www.electronicshub.org/artificial-neural-networks-ann/
5	https://www.tutorialspoint.com/data_mining/
6	https://towardsdatascience.com/the-fp-growth-algorithm

List of Experiments:

1. Basics of Prolog Programming language.

2. Write simple fact for following:
a. Ram likes mango.
b. Seema is a girl.
c. Bill likes Cindy.
d. Rose is red.
e. John owns gold.

3. Write predicates one converts centigrade temperatures to Fahrenheit, the other checks if a temperature is below freezing.

4. Write a program to implement factorial fibonacci of a given number.

5. Write a program to solve 8-Queens problem.

6. Solve any problem using depth first search.
7. Solve any problem using best first search.
8. Solve 8- puzzle problem using best first search
9. Solve Traveling Salesman problem.
10. Write a program to solve the Monkey Banana problem.
11. Write a program to solve water jug problem using LISP.
12. Write a program in prolog to solve Tower of Hanoi